Clustering

- The process of grouping any kind of data based on the similarity in their features, automatically, without human expertise, is called **clustering**. It is a type of **unsupervised learning**.

- Intuitively, clustering is dividing a population into groups such that the points in one group are similar to each other. Each group is called a **cluster**.

- Since there is no ground truth data and nothing to compare with, we decide if a cluster is good or bad if it simply makes some **business sense.**

Some advanced applications of clustering:

➔ Validation of domain knowledge/human opinion.

➔ Search algorithms and recommendation systems.

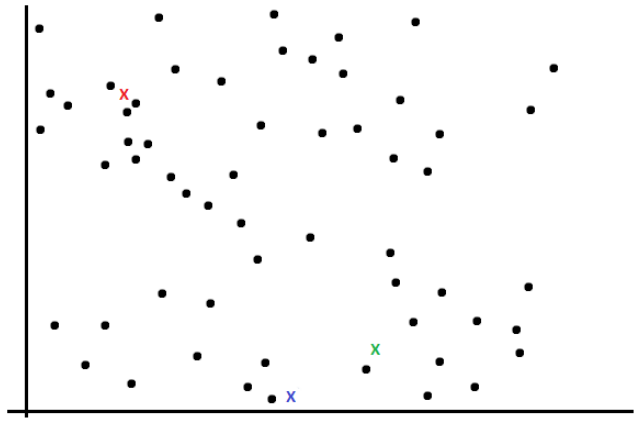
➔ Feature creation.

➔ Clustering with supervised learning to improve model accuracy.

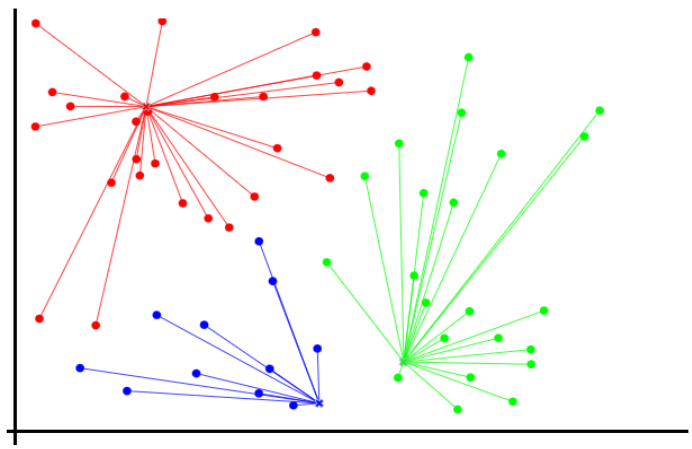
Lloyd's algorithm (K-means algorithm)

- This algorithm is used to cope with the problem of updating the centers.
- It has 5 basic steps:

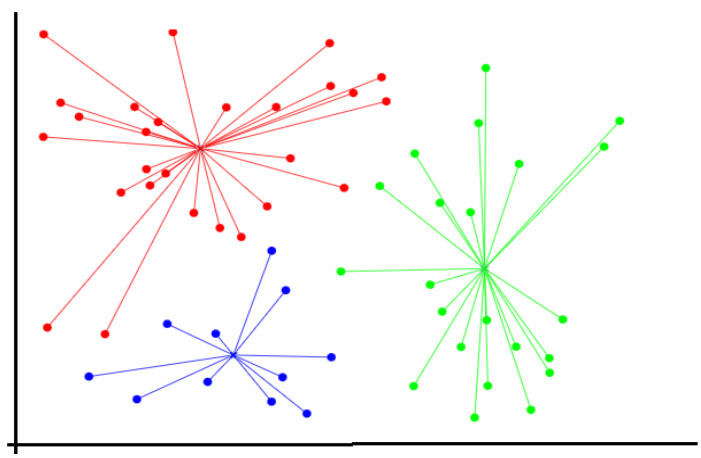
➔ Randomly initialize **k** centers.



➔ Assign points to these centers to get the clusters.



➔ Find the **centroids** of these clusters.



➔ Reassign the points to the centers to create the new clusters.

➔ Repeat until the center of new clusters stops changing their positions.

Within-cluster sum of squares (WCSS)

- The within-cluster sum of squares is a measure of the variability of the data points within each cluster. It is given as:

$$wcss = \sum_{i=1}^{k} \sum_{j=1}^{m_i} (x_{ij} - c_i)^2$$

where $\mathbf{x_{ij}}$ is the $\mathbf{j^{th}}$ point belonging to the $\mathbf{i^{th}}$ cluster and $\mathbf{m_i}$ is the number of points in the $\mathbf{i^{th}}$ cluster.

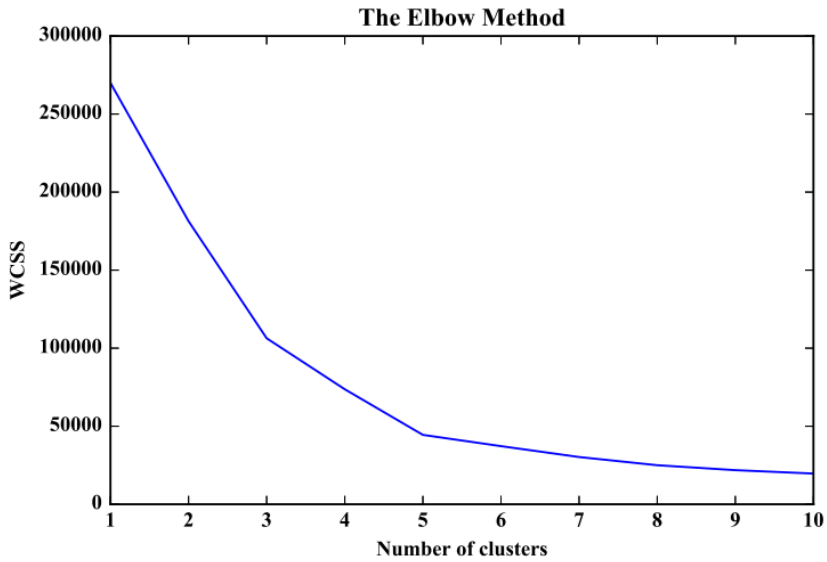
- A **variation** of the above formula can be as follows:

$$wcss = \sum_{i=1}^{k} \sum_{j=1}^{m_i} d(x_{ij}, c_i)$$

where, $\mathbf{d(x_{ij}, c_i)}$ is representing a distance function that is calculating the distance between the point $\mathbf{x_{ij}}$ and the centroid $\mathbf{c_i}$ of the cluster.

Elbow method

- It is a method to determine the optimal number of clusters (**k**) for k-means clustering.
- We perform the k-means clustering for a range of values of **k** and for each iteration, we calculate the value of the WCSS metric.
- When the value of WCSS is plotted against a range of **k** values, we get a plot looking like an elbow.



- We can clearly see that the WCSS value decreases as the number of clusters (k) increases.

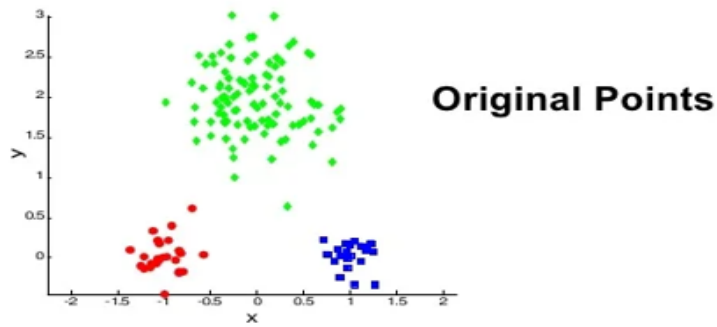
- At some point on the graph, there is a sharp change in the slope (k = 5) after which the change in slope is very small. The k value corresponding to this point is the optimal K value or an **optimal** number of clusters.

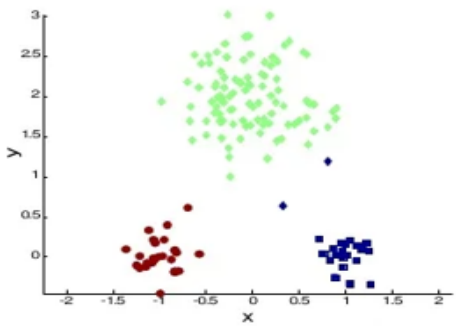
- If we do not get a sharp change in the slope of the elbow plot while using the WCSS metric on the y-axis, we can try using the **Silhouette score** to get significant results or to get confidence in our decision.

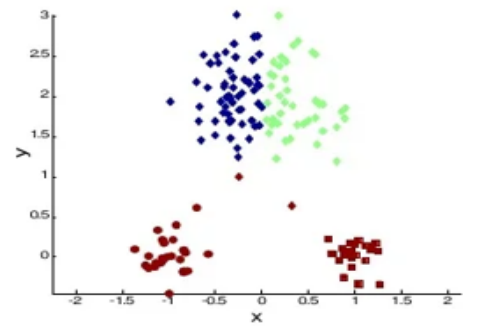
Drawbacks of the K-means

1. We may get bad results sometimes while using the k-means algorithm because of the **random initialization** of the centroids.

   For example







2. The k-means algorithm may not give the best results for data where the clusters are of varying size or density.

3. The number of clusters (k) is not defined.
4. It does not work well with non-globular clusters.

K-means++ algorithm

- To overcome the drawback due to the random initialization of centroids in K-means clustering, we use K-means++. It is smarter to initialize the centroids in order to improve the clustering algorithm.

- The steps involved in the initialization of centroids are:

- ➔ Select the first centroid randomly from the data points.
- ➔ Choose the next center as the farthest point from the first center.
- ➔ The next center would be a data point farthest from both the first and second centers.

- Repeat steps 2 and 3 until **k** centroids have been sampled.

- If there are **outliers** in our data, then instead of choosing them as centroid, we can choose the farthest point as the centroid with a **probability proportional to the distance.** This is the implementation that sklearn follows by default..

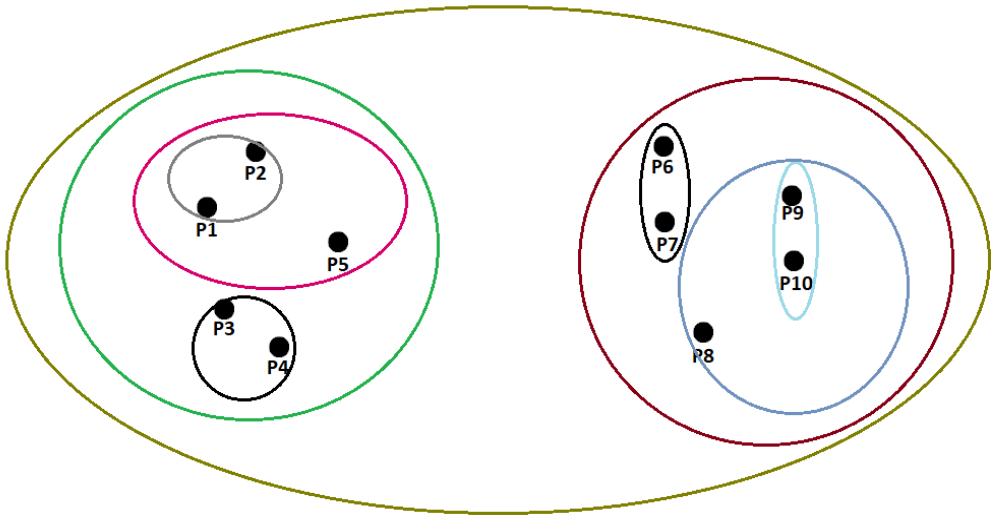
Hierarchical clustering

- It is an unsupervised clustering algorithm in which the hierarchy of clusters is developed in the form of a tree, and this tree-shaped structure is known as the **dendrogram.**

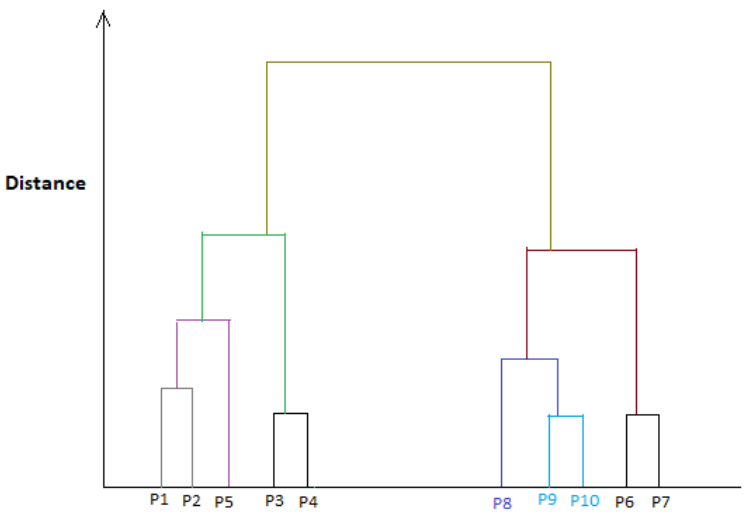
Steps of Hierarchical clustering:

- ➔ Compute the distance between all the data points. We can choose any distance measure from Euclidean, Manhattan, Hamming Distance, etc
- ➔ Merge the two points having minimum distance between them.
- ➔ Treat the sub-clusters as points and repeat till only one cluster remains.
- ➔ Finally, develop the dendrogram to divide the clusters as per the problem.

For example,



The associated **dendrogram** is:



Distance between two clusters:

- We observe that the distance between two clusters needs to be calculated in order to merge two clusters. Therefore, some linkage needs to be formed between these clusters.

- The common type of linkages that can be formed between the two clusters are:

1. Complete linkage: It is the **maximum** distance between the two farthest points of different clusters.
2. Single linkage: It is the **minimum** distance between the two closest points of different clusters.
3. Centroid linkage: The distance between the centroids of the clusters is calculated.
4. Ward's distance/linkage: The Ward's distance between two clusters c_1 and c_2 is given as:

$$W(c_1 + c_2) = WCSS(c_1 + c_2) - WCSS(c_1) - WCSS(c_2)$$

where WCSS is the **within-cluster sum of squares**.

Advantages of Hierarchical clustering:

- ➔ There is no need to pre-decide the k value. i.e. no. of clusters.
- ➔ Good for data visualization providing hierarchical relation between the clusters. The hierarchy may actually be present in the real world.

Disadvantages of Hierarchical clustering:

- ➔ This method of clustering is sensitive to the choice of linkage function.
- ➔ Hierarchical clustering can only be used for analysis. We cannot do inference using it. If we get a new data point, we need to re-run the entire algorithm with all the existing points and reanalyze.

DBSCAN (Density-based spatial clustering application with noise)

- The fundamental principle is that each point in a cluster must have at least a certain number of neighbors within a given radius.

- The density threshold is defined by two parameters: the radius of the neighborhood (**eps**) and the minimum number of neighbors (**minPts**) within the radius of the neighborhood.

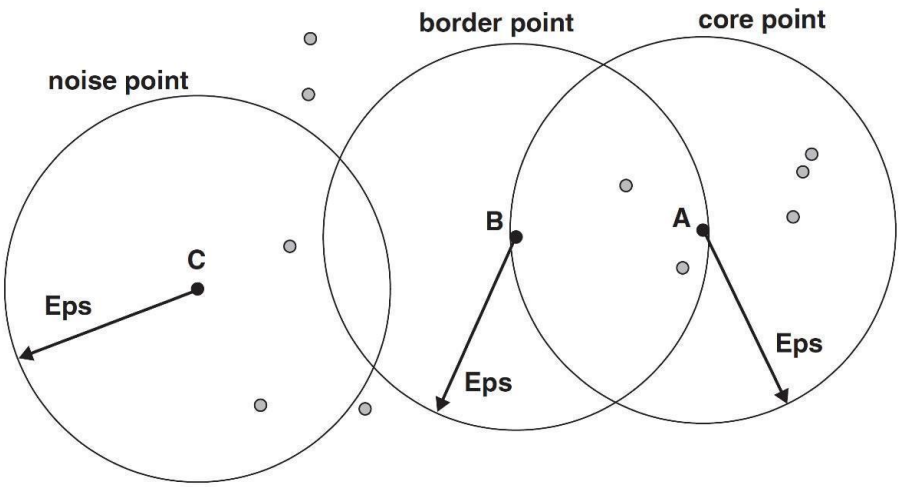
- There are three types of points in our data:

1. Core point: This is a point that has at least *minPts* points within distance *eps* from itself.

2. Border point: This is a point that has at least one Core point at a distance *eps and it is not a core point*.

3. Noise point: This is a point that is neither a Core nor a Border. And it has less than *minPts* points within distance *eps* from itself.

For e.g., with a value **minPts= 7**,



- The DBSCAN clustering algorithm can be summarized in the following steps:
  - ➔ Pick a random point from the dataset and if it is a **core point**, assign it to cluster 1 and mark it as visited.
  - ➔ If not, the point is marked as noise. Assign all the points within the neighborhood of the initial point to cluster 1. If these new points are also core points, assign their neighborhood point to cluster 1.
  - ➔ Next, randomly choose another point that has not been visited and repeat step 1 with it.
  - ➔ Stop when all the points are visited.

Deciding *epsilon*:

- One way to estimate the initial value of epsilon is:
  - ➔ Calculate the distance between each point.
  - ➔ Plot the histogram of those distances.
  - ➔ We may get two peaks, the initial value of epsilon may be one of the distance values between these two peaks.

Advantages of DBSCAN:

- ➔ It is able to find arbitrarily shaped clusters and clusters with noise (i.e. outliers).
- ➔ There is no need to decide ‘k’ (no. of clusters).

Disadvantages of DBSCAN:

- ➔ It does not work well with sparse points (high dimensional data).
- ➔ While DBSCAN is great at separating high-density clusters from low-density clusters, DBSCAN struggles with clusters of similar density.

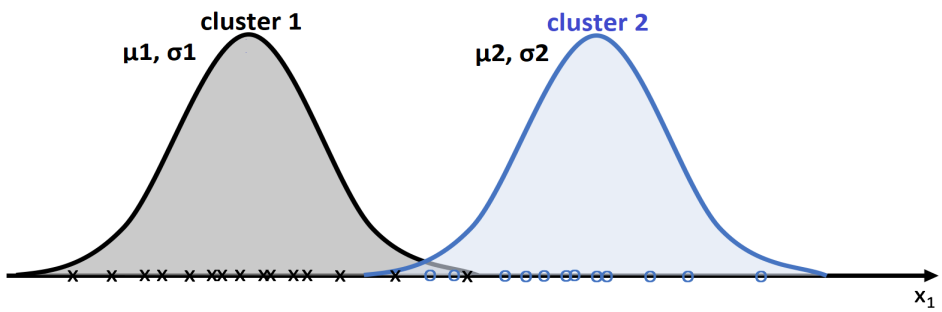
Soft clustering with Gaussian mixture model

- In soft clustering, a probability of each data point being in different clusters is assigned instead of putting each data point completely into a separate cluster.

- Gaussian Mixture Models are probabilistic models and use the soft clustering approach for distributing the points in different clusters.

- We can represent the clusters in our data using the **Gaussian distributions**.

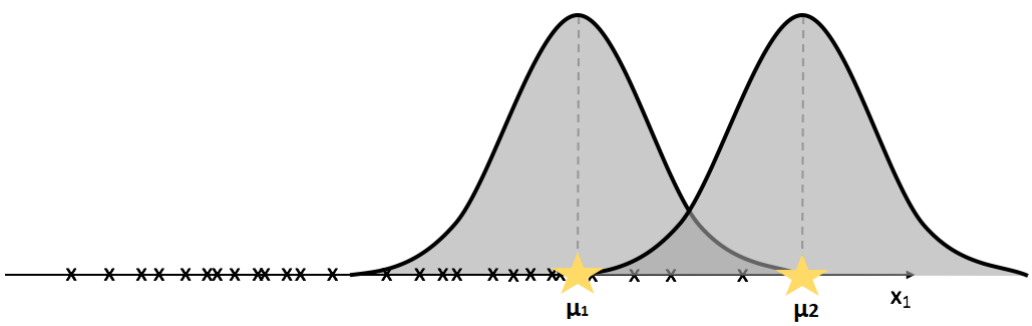
- For example, in the below given 1-Dimensional data, each cluster is represented using a Gaussian distribution.



1-D Example of GMM

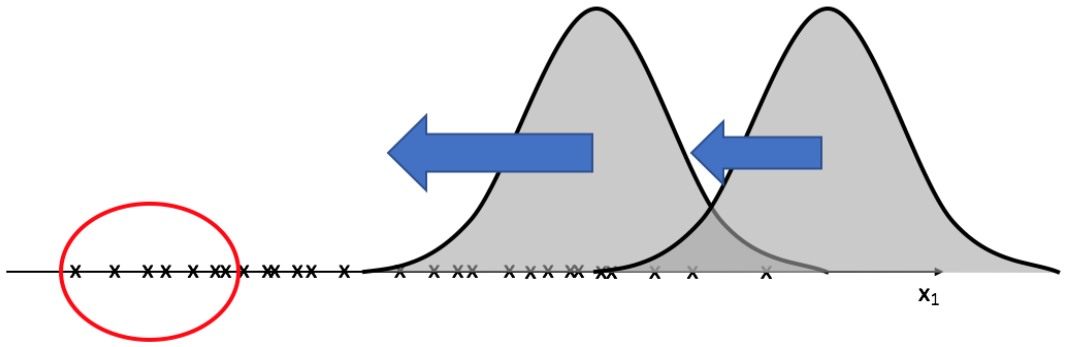
GMM algorithm:

➔ Randomly initialize μ and σ.



➔ Update μ and σ.

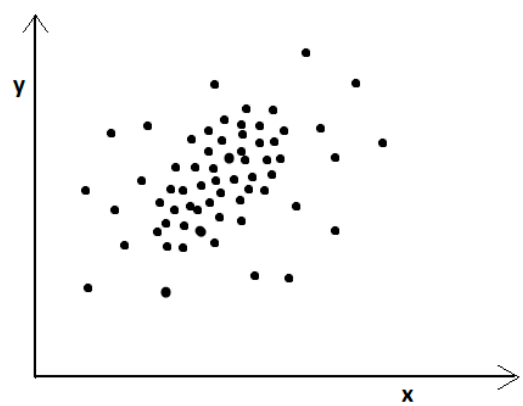
- Calc the probability of each point belonging to each gaussian.
- Assign all the points which have the highest probability to their respective Gaussians.
- Now, compute new μ for each gaussian as the mean of the points belonging to that gaussian (similar to Lloyd's algorithm)
- Similarly, update the variance.



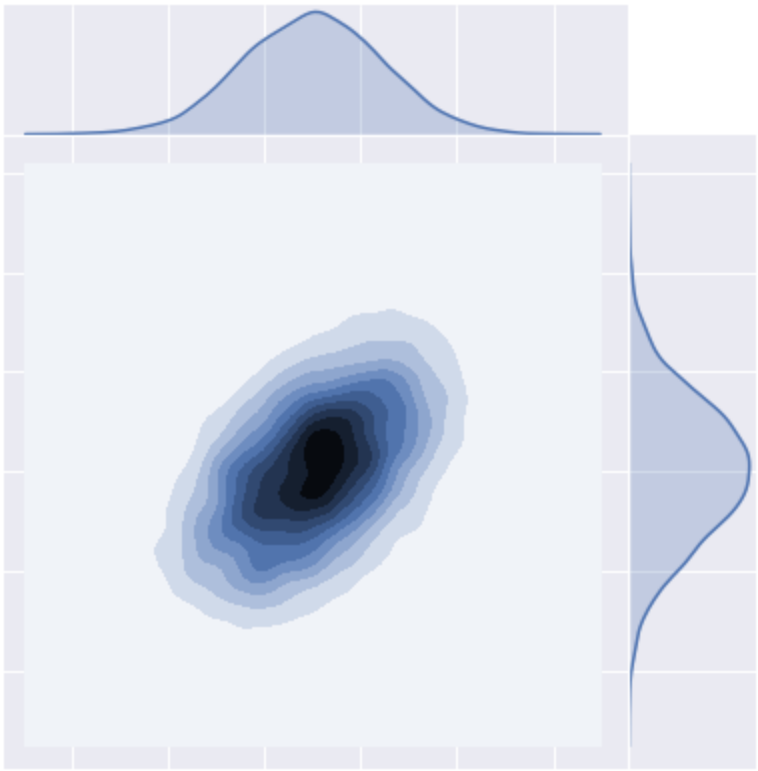
➔ After multiple updates, we will have tightly fitting Gaussian distributions representing different clusters.

2-Dimensional Gaussians:

- Let's consider two variables **x** and **y** with the following 2D scatter plot.



- The corresponding Multivariate Gaussian distribution would be:



- The part where the color is dark has more density of points and vice versa.

- The above multivariate Gaussian distribution is defined by five parameters:

    1. μ_x - the mean of x-coordinate.
    2. μ_y - mean of y-coordinate
    3. σ^2_x - variance in x direction.

    4. σ^2_y - variance in the y direction.
    5. **Cov(x,y)** - Covariance between x and y.

- The above ideas can be applied to 2D and ND gaussian, and this is how the GMM algorithm works.